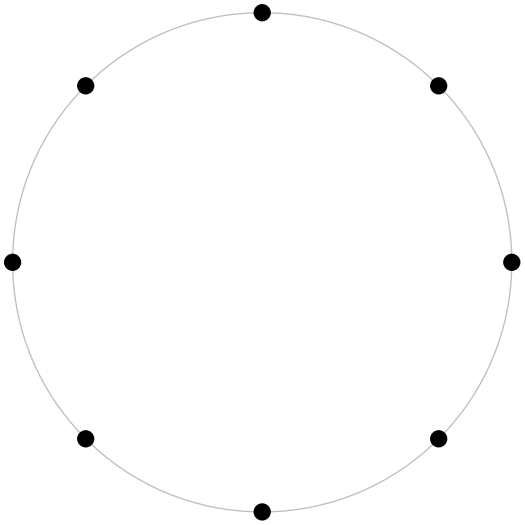
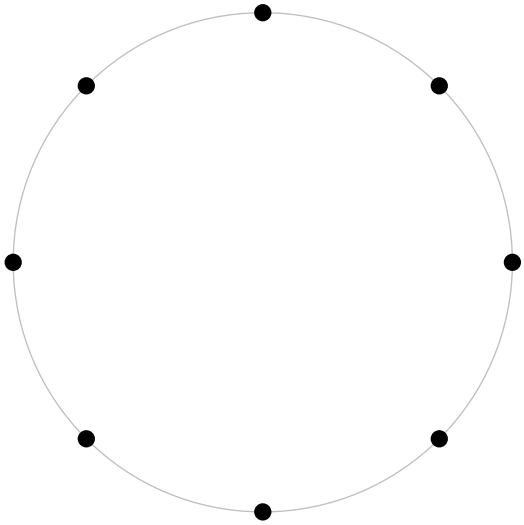
To draw hop 3 on a ring of dots, start at any dot and draw a straight line to the dot 3 places clockwise. Keep hopping 3 until you are back where you started. If some dots still have no line, start again at one of them, and stop when every dot has a line.

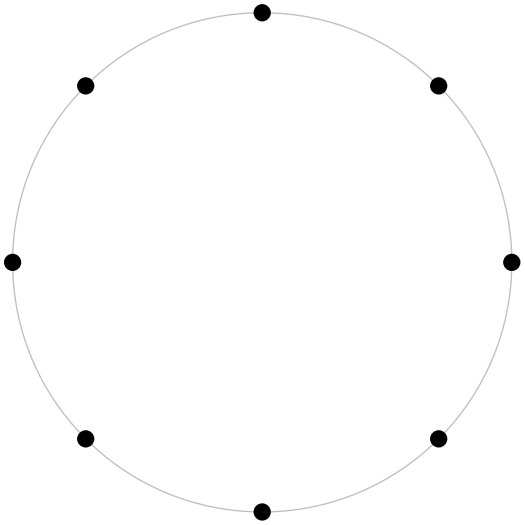
Problem 1: Draw hop 1, hop 2, hop 3 and hop 4 on these rings of 8 dots. Write how many times you started in the box under each drawing.



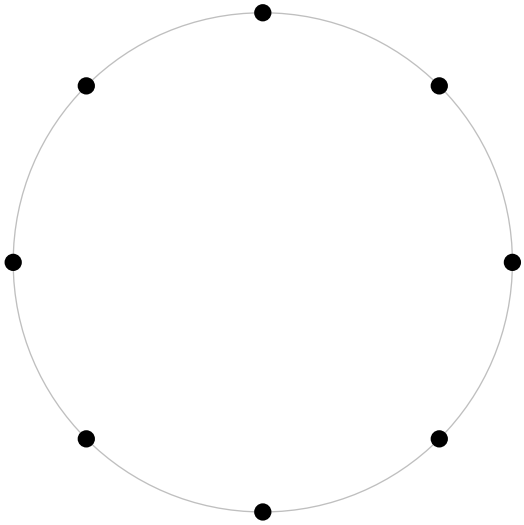
hop 1 ☐



hop 2 ☐

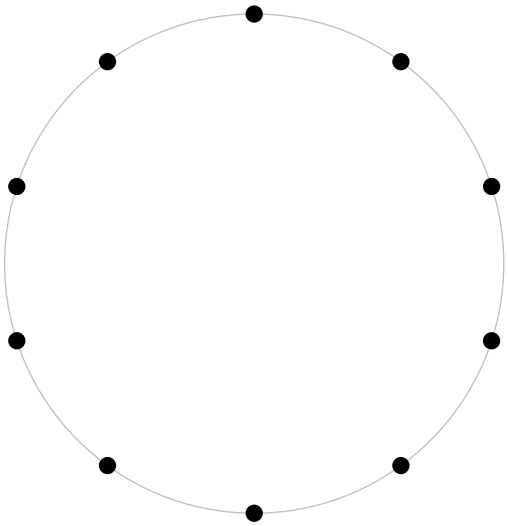


hop 3 ☐

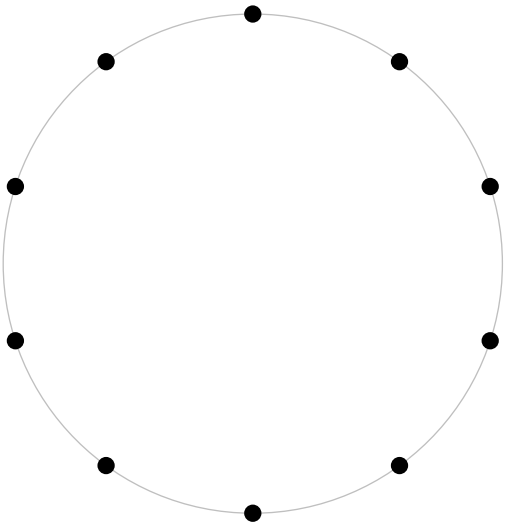


hop 4 ☐

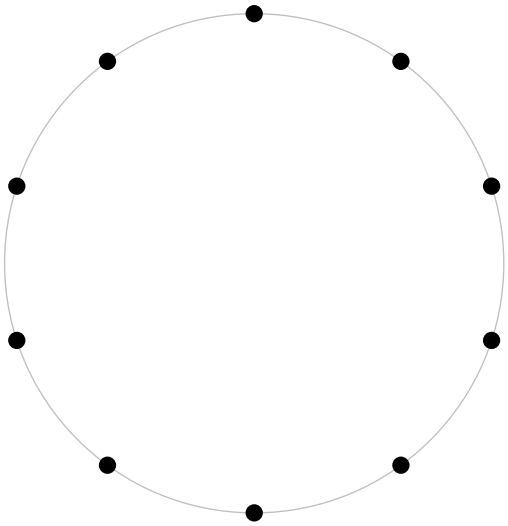
Problem 2: Draw hop 2, hop 3, hop 4 and hop 5 on these rings of 10 dots. Write how many times you started in the box under each drawing.



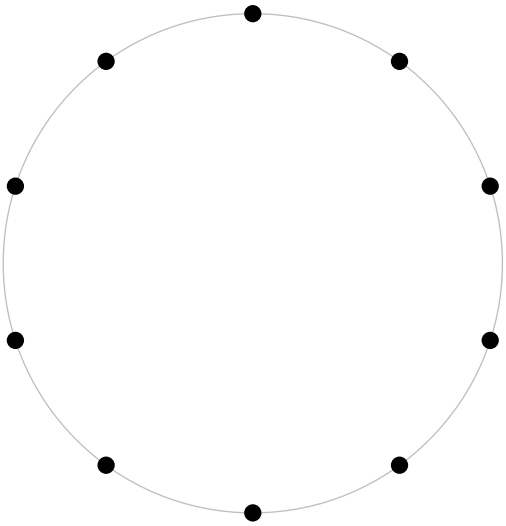
hop 2 ☐



hop 3 ☐

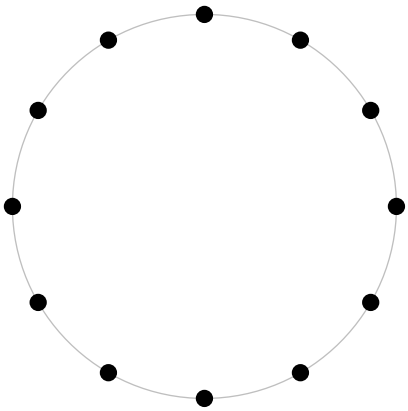


hop 4 ☐



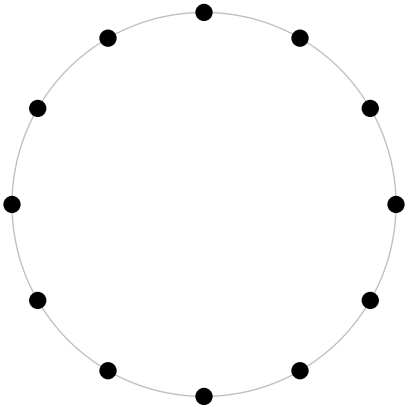
hop 5 ☐

Problem 3: Before you draw each hop on these rings of 12 dots, guess how many times you will start and write your guess. Then draw it and write how many times you did start.



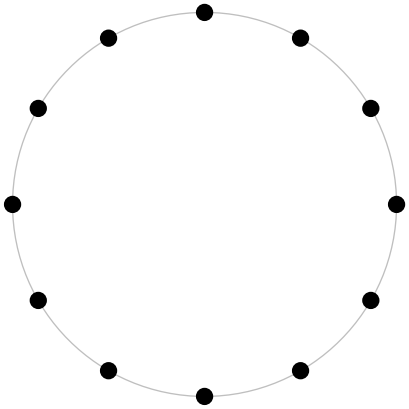
hop 2

guess ☐ started ☐



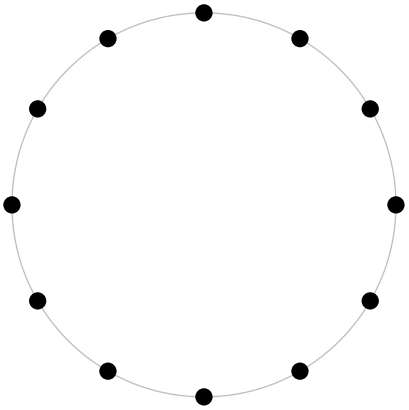
hop 3

guess ☐ started ☐



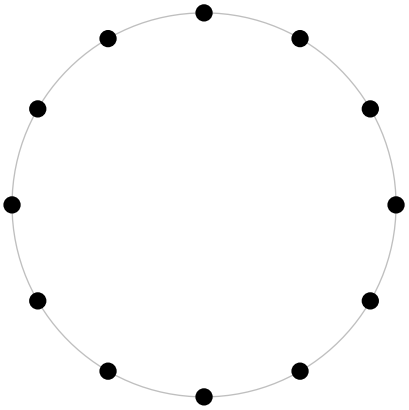
hop 4

guess ☐ started ☐



hop 5

guess ☐ started ☐



hop 6

guess ☐ started ☐

Problem 4: Cut out the two wheels on the last two pages and join them through the middle dots with a brass fastener. Setting the wheel to D means turning it until the inner A is next to the outer D. Each letter of a message on the inner wheel goes into code as the outer letter next to it, so with the wheel set to D, CAT goes into code as FDW. Set your wheel to D and find what each of these codes came from.

V W D U

__ __ __ __

I R A

__ __ __

Z K H H O

__ __ __ __ __

S H Q F L O

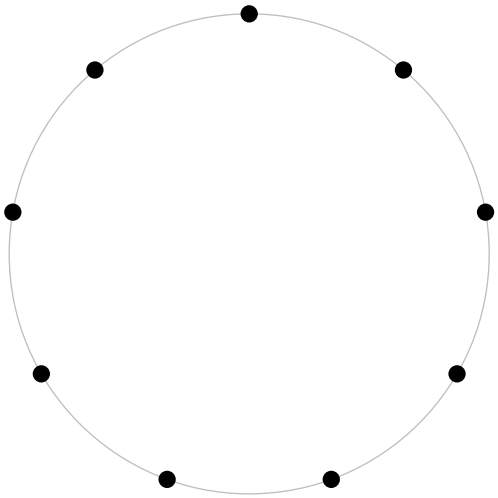
__ __ __ __ __ __

L F D Q G U D Z D V W D U

__ __ __ __ __ __ __ __ __ __ __ __ __

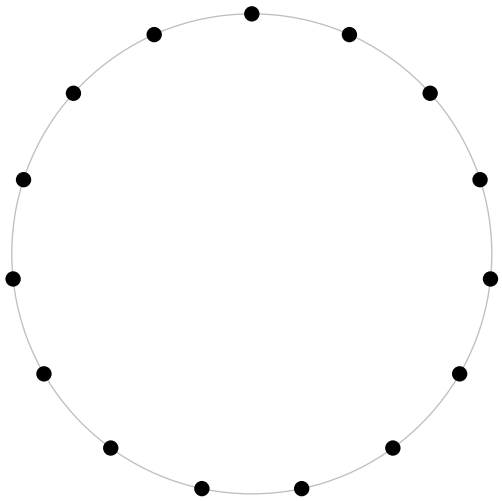
Problem 5: Write a message of two or three words for your partner, and put it into code with a setting you choose. Swap codes and settings with your partner, and find your partner's message.

Problem 6: Before you draw each of these, guess how many times you will start and write your guess. Then draw it and write how many times you did start.



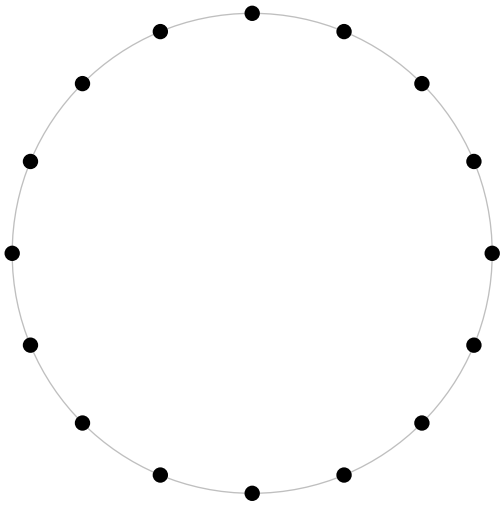
9 dots, hop 2

guess ☐ started ☐



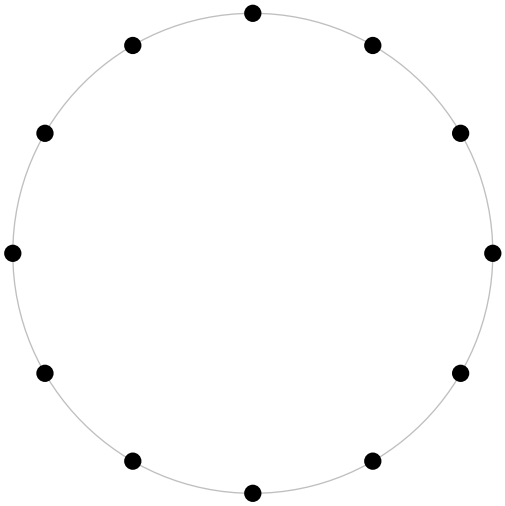
15 dots, hop 5

guess ☐ started ☐



16 dots, hop 6

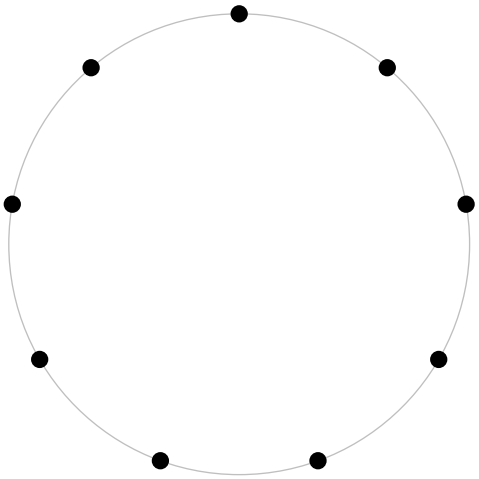
guess ☐ started ☐



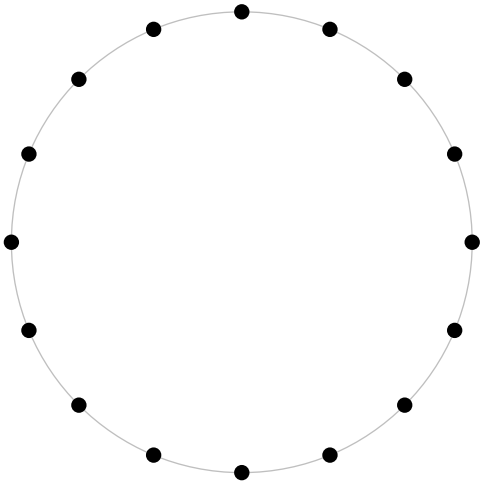
12 dots, hop 8

guess ☐ started ☐

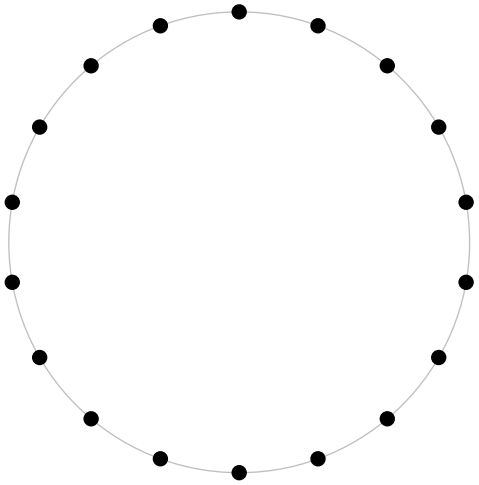
Problem 7: On each of these rings, find a hop bigger than 3 that makes you start exactly 3 times, and draw it. If a ring has no such hop, explain why.



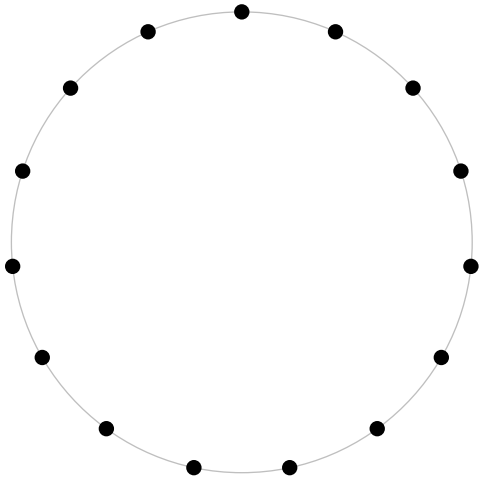
9 dots, hop _____



16 dots, hop _____



18 dots, hop _____



15 dots, hop _____

Problem 8: These messages were put into code with settings nobody wrote down. Find the messages.

H Z A H Y J H U O H C L A L U

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __

W V P U A Z

__ __ __ __ __ __

C U U J C U R O J X U R Y W

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __

J H U U

__ __ __ __

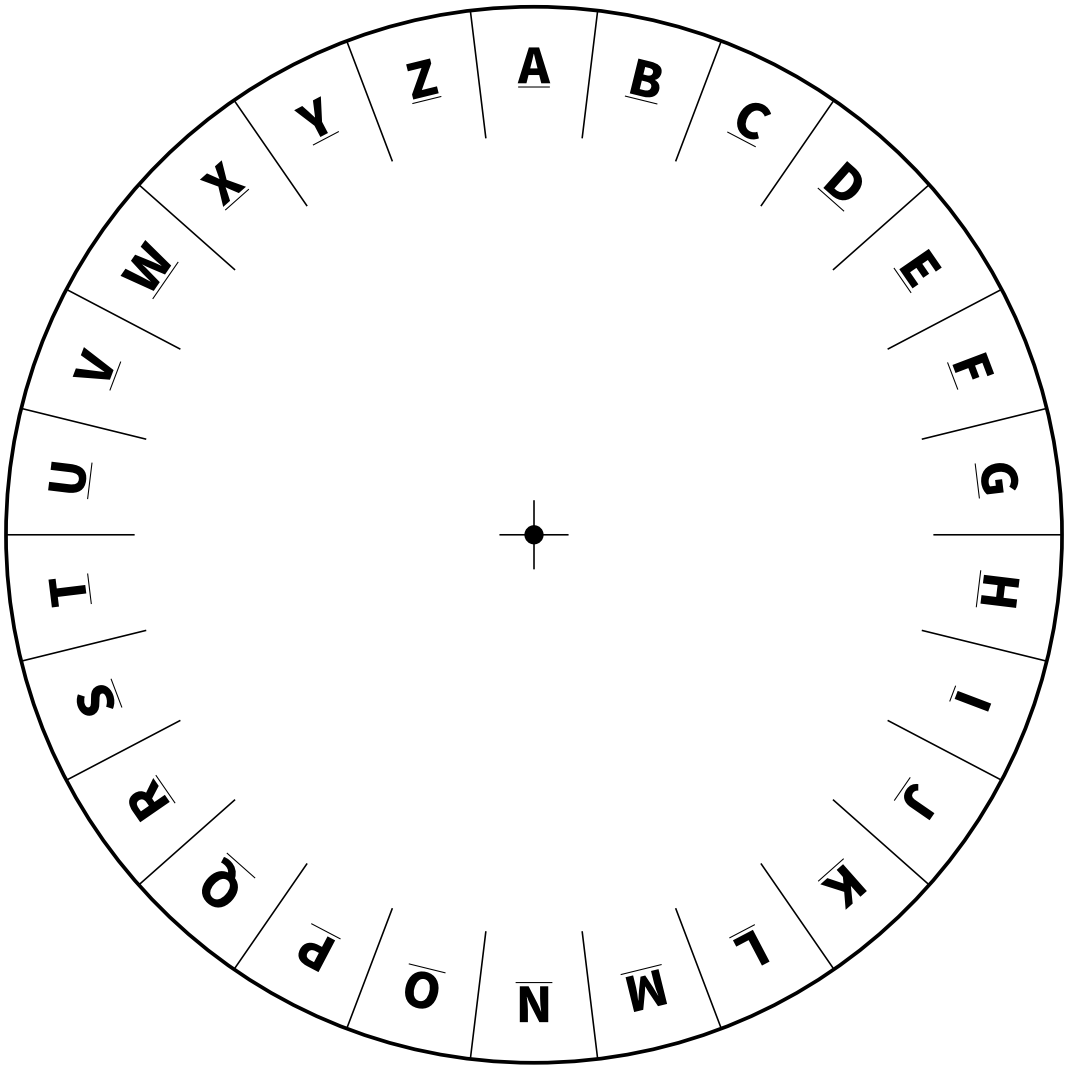
Problem 9: Sam put a word into code. Then he put that code into code again with a second setting, and he got his word back. Find the second setting for each of these first settings.

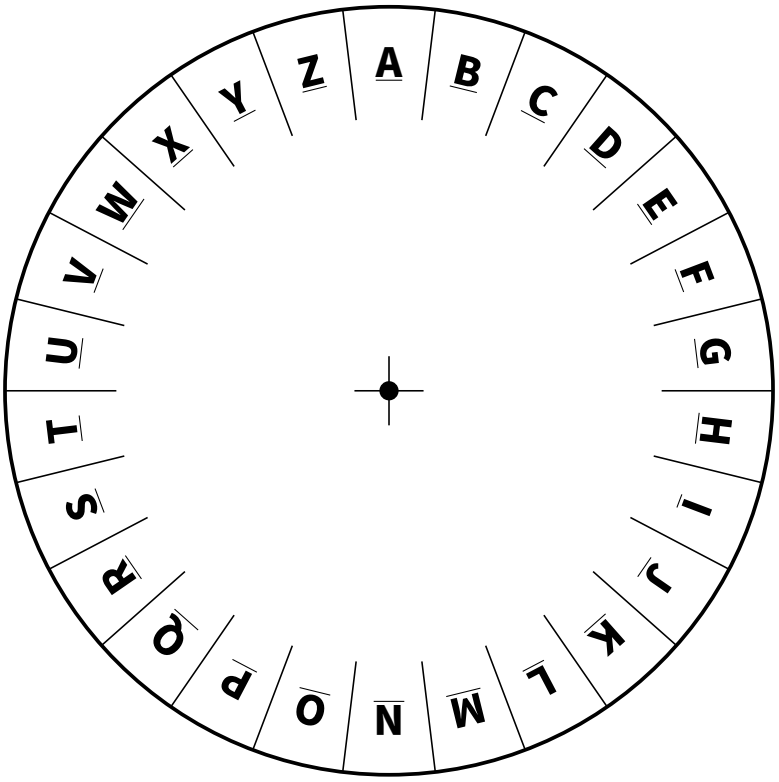
first setting D second setting ☐

first setting H second setting ☐

first setting N second setting ☐

first setting W second setting ☐


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z


A
B
C
D
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Z